Signed Pell Frames: A Nine-Auxiliary Ordinary Power Graph

Research iteration, 24 October 2026

1 A completed ordinary power graph with nine auxiliaries

Result and scope. For positive integers n, k and a natural m, the relation $m = n^k$ has the following ordinary polynomial graph with **nine auxiliary natural variables and exact degree** 20. More generally, r simultaneous positive-power requests use $4r + 5$ auxiliaries and exact degree $12r + 8$. The inputs are separate from the auxiliaries. This is a complete polynomial result, not a special-function interface, and does not claim a new fewest-variable consistency equation.

The common scale and common Pell-index certificate are inherited from the reviewed synchronized construction. The new reduction eliminates the common Pell base even though the resulting polynomial can be negative. Instead of introducing a sign witness, we prove that the entire negative branch is harmless. No historical minimality claim is made.

1.1 The explicit ordinary polynomial

Fix $r \geq 1$ and inputs $n_i, k_i \geq 1$, $m_i \geq 0$. Quantify only

$$z, U, v, q, b, \qquad y_i, h_i, s_i, t_i \quad (1 \leq i \leq r)$$

over $\mathbb{N}$. Define polynomial abbreviations

$$\begin{aligned}
&W = 2 + \sum_i (n_i + k_i + m_i), && Y_i = k_i + y_i, && D = \prod_i Y_i,\\
&V = D^2(v+1), && B = 1 + 4D(q+1), && A = B - Ub,\\
&T_i = 2An_i - n_i^2 - 1, && X_i = Y_i(A - n_i) + m_i + T_i h_i, &&\\
&S_i = X_i + Us_i, && R_i = k_i + 4Y_i t_i. &&
\end{aligned}$$

Set $N(a, x, y) = x^2 - (a^2 - 1)y^2 - 1$. The entire equation is

$$\boxed{\begin{aligned} E_r = {}& \left[A^2 - W^3(W+2)(z+1)^2 - 1\right]^2 + N(A, U, V)^2 \\ &+ \sum_{i=1}^{r} \left(N(A, X_i, Y_i)^2 + N(B, S_i, R_i)^2\right) = 0. \end{aligned}} \tag{1}$$

All products have the fixed length r, and every exponent in the displayed recipe is the numeral 2 or 3. In particular A, T_i, X_i are polynomial abbreviations, not additional variables or assumed positive quantities.

Theorem. For the stated input domain, $E_r = 0$ has a solution in its $4r + 5$ natural auxiliaries if and only if $m_i = n_i^{k_i}$ for every i.

1.2 The signed elimination argument

Suppose $E_r = 0$. All residuals vanish. The scale equation yields $|A| > W^2$, while $Y_i \geq k_i \geq 1$ and $n_i, k_i, m_i \leq W$. The potentially dangerous case is $A = -\alpha < 0$. Put $\tau_i = 2\alpha n_i + n_i^2 + 1 > m_i$. Then

$$X_i = -(\alpha + n_i)Y_i + m_i - \tau_i h_i < 0.$$

Writing $x_i = -X_i > 0$, its norm gives $x_i \leq \alpha Y_i$. Its definition therefore implies

$$n_i Y_i + \tau_i h_i \leq m_i, \qquad Y_i \leq m_i \leq W < \alpha.$$

If $Y_i > 1$, the factorization

$$(\alpha Y_i - x_i)(\alpha Y_i + x_i) = Y_i^2 - 1$$

has positive integer factors, with the second greater than αY_i. Thus $Y_i^2 - 1 > \alpha Y_i$, contradicting $Y_i < \alpha$. Hence $Y_i = 1$, $k_i = 1$, $x_i = \alpha$, and $m_i = n_i + \tau_i h_i$. Since $\tau_i > m_i$, we obtain $h_i = 0$ and $m_i = n_i = n_i^{k_i}$.

This proves soundness of the complete negative branch without using an index theorem, changing a congruence sign, or adding a positivity witness. The branch is real: the checker includes explicit negative-A zeros. If any requested exponent is at least two, that branch is impossible.

1.3 Positive soundness

For $A > 0$ we have $A > W^2$, so $A \geq 2$, $A > n_i$, and

$$T_i \geq 2A - 2 > W \geq m_i.$$

All coordinates needed by the standard Pell-index criterion are now nonnegative, and $Y_i, V > 0$, $B \geq 5$. The definitions supply

$$Y_i^2 \mid V, \quad B \equiv 1 \pmod{4Y_i}, \quad B \equiv A \pmod{U}, \quad S_i \equiv X_i \pmod{U}, \quad R_i \equiv k_i \pmod{4Y_i}.$$

Together with the three norms and $k_i \leq Y_i$, the exact Matiyasevich criterion identifies $X_i = \chi_A(k_i)$ and $Y_i = \psi_A(k_i)$, where

$$\chi_a(j) + \psi_a(j)\sqrt{a^2 - 1} = (a + \sqrt{a^2 - 1})^j.$$

The imported criterion includes its signed-index analysis; it is the statement exposed by Mathlib's `Pell.matiyasevic` and the repository's `PellRepresentation.criterion_iff_pell`.

The scale norm is at Pell base $W + 1$, with second coordinate $W(z + 1) > 0$. Classification gives an index $j > 0$. Modulo W, its recurrence gives $\psi_{W+1}(j) \equiv j$, so $W \mid j$ and $j \geq W$. Consequently

$$A \geq (W + 1)^j > W^W \geq n_i^{k_i}, \qquad T_i > n_i^{k_i}.$$

For $H_j = \chi_A(j) - (A - n_i)\psi_A(j)$ the exact recurrence is

$$H_0 = 1, \qquad H_{j+1} = n_i H_j + T_i \psi_A(j).$$

Thus $H_{k_i} \equiv n_i^{k_i} \pmod{T_i}$. The definition of X_i also gives $H_{k_i} = m_i + T_i h_i$. Both proposed and true outputs lie in $[0, T_i)$, establishing equality.

1.4 Simultaneous completeness

Assume all requested powers are true, and calculate W first. Choose

$$A = \chi_{W+1}(W), \quad z = \psi_{W+1}(W)/W - 1, \quad X_i = \chi_A(k_i), \quad Y_i = \psi_A(k_i).$$

The recurrence modulo W makes z natural. Set $y_i = Y_i - k_i$ and

$$h_i = \sum_{j=0}^{k_i-1} n_i^{k_i-1-j} \psi_A(j) \geq 0.$$

The recurrence above proves the required expression for X_i. These powers and finite sums occur in the witness proof only, not in equation (1).

Put $D = \prod_i Y_i$. In the finite ring

$$(\mathbb{Z}/4D^2\mathbb{Z})[s]/(s^2 - (A^2 - 1)),$$

the element $A + s$ is a unit with inverse $A - s$. A positive finite-group order supplies $U = \chi_A(\ell) \equiv 1 \pmod{4D^2}$ and $V = \psi_A(\ell) \equiv 0 \pmod{4D^2}$, with $V > 0$. Hence $v = V/D^2 - 1 \geq 0$, and $\gcd(U, 4D) = 1$. CRT gives

$$B \equiv A \pmod{U}, \qquad B \equiv 1 \pmod{4D}.$$

Increase B by a multiple of $4DU$ until $B > A$ and $B > 1$, and set

$$b = (B - A)/U, \qquad q = (B - 1)/(4D) - 1.$$

Finally take $S_i = \chi_B(k_i)$, $R_i = \psi_B(k_i)$ and

$$s_i = (S_i - X_i)/U, \qquad t_i = (R_i - k_i)/(4Y_i).$$

Polynomial dependence on the base proves these quotients integral. Monotonicity for $B > A$ proves them nonnegative. Every residual vanishes, and $A = B - Ub$ restores exactly the chosen scale. No source consistency, prime-distribution assertion, or circular existence argument is used.

1.5 Honest counts and the tiling boundary

With independent input ports,

$$\deg V = 2r + 1, \quad \deg A \leq r + 1, \quad \deg X_i, \deg S_i \leq r + 3.$$

The auxiliary norm has degree $6r + 4$, while each local norm has degree at most $2r + 6$. Its leading component contains $-16q^2v^2(\prod_i Y_i)^6$, so the final degree is exactly $12r + 8$. For one request, the delivered DAG has 12 total variables, 9 auxiliaries, 4 residuals and degree 20. For two requests it has 19 total variables, 13 auxiliaries, 6 residuals and degree 32.

Retaining A as an independent natural variable and adding the residual $B - A - Ub$ gives a lower-degree sibling. For two requests this uses 14 auxiliaries and degree at most 24. This is a coordinate–degree tradeoff, not universal dominance.

The preceding two-power strict-boundary tiling interface has 8 exposed coordinates. Literal substitution now gives an ordinary polynomial with 21 natural variables, or 22 for the lower-degree sibling. For k tiles, the complete substituted degree bounds are respectively $1728k + 8$ and $1152k + 8$; they are *not* the standalone graph degrees 32 and 24. The complete four-corner fixture has bounds 6920 and 4616, including every fixed-power packed-circuit operation.

Not a new consistency endpoint. A strict-boundary tile set correctly initialized and calibrated to an actual ZFC proof checker is still required. The raw 943-tile list cannot simply be substituted: it contains an all-zero tile, admitting a trivial one-cell rectangle under this generic boundary condition. We therefore claim a completed power-graph lemma and generic tiling compiler, not a 21-variable Con(ZFC) polynomial or an improvement to the calibrated 9/10-positive and 11-integer frontiers.

Validation. The packet includes complete polynomial DAGs, separately transcribed evaluators, 1500 modular whole-polynomial comparisons, 16 full positive- and negative-base exponent-one witnesses, genuinely different-index CRT fixtures, 479080 finite negative-base controls, exact-degree certificates, and corruption tests. Normal and optimized Python runs pass using explicit exceptions rather than disabled assertions. Source hashes are pinned. These tests supplement the universal proof; no new Lean verification, encoded internal ZFC proof object, or proof of consistency is claimed.

Independent ordinary review. The separate review accepts the signed branch, positive branch, common-scale bound and simultaneous CRT construction. Its independent symbolic checker compares every generic port and residual with the literal emitted DAG and confirms exact degrees 20, 32 and 24 for the three generic artifacts. It does not certify the separate tiling-source calibration.

Sources and reproduction. The copied Mathlib source is pinned to commit `4dc43842a11f` (full hash in the source pin); its relevant Pell-index theorem is included with its license. The exact repository wrapper and inherited shared-frame proofs are separately authenticated in `SOURCE_PINS.json`. No incoming archive was changed.

The complete recipe can be rebuilt with `python build.py -check`; the separate checker is `python check.py`. Repeat both with `python -O` to verify that no crucial test relies on an assertion which optimized mode can remove. The numerical input-domain restrictions remain part of the theorem in every mode.